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(54) Title of the invention : AI-BASED WORKFLOW AUTOMATION SYSTEM FOR INTELLIGENT BUSINESS PROCESS MANAGEMENT

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(57) Abstract :

The system (100) includes An AI-Based Workflow Automation System for Intelligent Business Process Management comprises: a data acquisition unit (102), a preprocessing module (104), a process modeling engine (106), a rule-based decision engine (108), a machine learning module (110), a workflow orchestration engine (112), a user interface module (114), and a reporting and analytics module (116). The data acquisition unit (102) collects data from various internal and external sources. The preprocessing module (104) cleanses and transforms the collected data for further processing. The process modeling engine (106) constructs dynamic models of business processes based on the preprocessed data. The rule-based decision engine (108) applies predefined business rules to guide process flows. The machine learning module (110) analyzes historical data to predict outcomes and optimize processes. The workflow orchestration engine (112) manages the execution of business processes across different systems. The user interface module (114) provides users with interactive dashboards and control panels. The reporting and analytics module (116) generates insights and performance metrics for decision-making. The invention enhances operational efficiency by automating complex workflows and facilitates informed decision-making through predictive analytics.

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